



that I learnt there and have carried back. Because India today is in the right position to take that big leap in manufacturing hardware as well, like we did in software.

Building an ecosystem

Scaling manufacturing needs an ecosystem. Other than everything a factory needs, a good ecosystem comprises investors, sellers, and traders too. These are extremely crucial to the entire manufacturing process.

The investors who have made money in manufacturing will tell you how you could run the company, where you could find the customers, important insights to avoid mistakes, etc. If they are missing from your ecosystem, then you could go wrong in many aspects of manufacturing. Why Indian software industry is prospering today is because there are many rich veterans to guide the industry, while the Indian hardware ecosystem needs to be built bottoms up.

In China, however, the government is the designer. They are the ones who allocate separate districts or separate segments. The traders' and investors' district is located in Futian. The product designers' district is in Nanshan. The manufacturing district is located in Bao an.

All of them are within Shenzhen

city, separated as clusters but very well connected to each other. It is a well-planned system with good incentives to follow the plan and set up offices in the respective clusters.

How did they make it happen?

The entire pool of traders and investors used to be in Hong Kong earlier, and they had all the money for trading Japanese and American products. But over time, because of Chinese government's incentives, they started moving to China's Luohu district, as it was closer to the borders of Hong Kong. It resulted in the eventual shift of the investors to Luohu itself.

The Hong Kong traders were very experienced and passed on all the demand information to Shenzhen. This is one of the reasons why investors and traders are so crucial for this ecosystem. The government of China incentivised the investors or expats who shifted base to China from Hong Kong and Taiwan, where the experienced were originally settled.

Narrowed focus

I have been to Mauritius before, which also does a lot of manufacturing. The major difference that I figured in Mauritius as compared to Shenzhen was that a factory in Mauritius was responsible for every-

thing from production to packaging. Whereas in Shenzhen every process was broken down and was done separately by different factories.

Mauritius model is good for standalone factories but cannot scale up as a hub of manufacturing like China's. So, to make an ecosystem efficient, it is important to cluster small manufacturing units and keep the entire ecosystem together with many specialist factories having narrowed focus. There can be factories that just handle packaging or factories that take care of the accessories. This can be done in India with the help of the government or through private manufacturing Special Economic Zones (SEZs).

Skilled labour

India has a huge number of people who are working as labourers. But what matters is whether they are skilled enough to take up manufacturing jobs, which should be possible when the focus is narrowed.

Today, an average Chinese worker draws a much higher salary than an average Indian worker. But the labour in China is relatively well skilled. People who are interested in manufacturing should be investing in labour skill-sets as well in order to obtain better productivity.

Finally, we can conclude that India has its own problems and has a long way to go when it comes to becoming a key manufacturing destination. India needs traders, investors, strong associations, and like-minded people to take initiatives collectively. Since India is a democracy, we have the freedom to come together and pool our ideas to ultimately drive this our way than just wait for government to do it for us, as in the case of China. **EFY**

This article is based on a tech talk at India Electronics Week by Nikhil Bhaskaran, Founder-Director, Brainy Pi. It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY, who has a knack for open source exploration and research